



GET IT ON
Google Play

TITLE:

GOOGLE PLAY MARKET ANALYSIS &
INSIGHTS

SUBTITLE:

DATA-DRIVEN EVALUATION OF APP
PERFORMANCE, PRICING IMPACT, AND
USER SENTIMENT

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COVERS

- Problem Statement
 - Hypothesis
 - Data & Tools
 - Exploratory Data Analysis (EDA)
 - Free vs Paid App Insights
 - Sentiment Analysis
 - Key Findings
 - Recommendations
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PROBLEM STATEMENT

The Google Play marketplace contains thousands of apps across diverse categories. Understanding what drives app success—installs, ratings, price, and user sentiment—remains unclear.

This project aims to uncover patterns that explain why some apps perform significantly better than others



HYPOTHESIS



We hypothesize that app success in the Google Play Store is driven by four key factors:



Category saturation – categories with more competition may have lower installs per app.



Pricing strategy – Free apps will achieve significantly higher installs than Paid apps.



User satisfaction (ratings) – Higher ratings should strongly correlate with higher installs.



User sentiment trends – Positive sentiment in reviews should align with higher ratings.



Additionally, we expect:



Free apps to dominate installs,



Paid apps to receive stronger ratings due to user expectations.



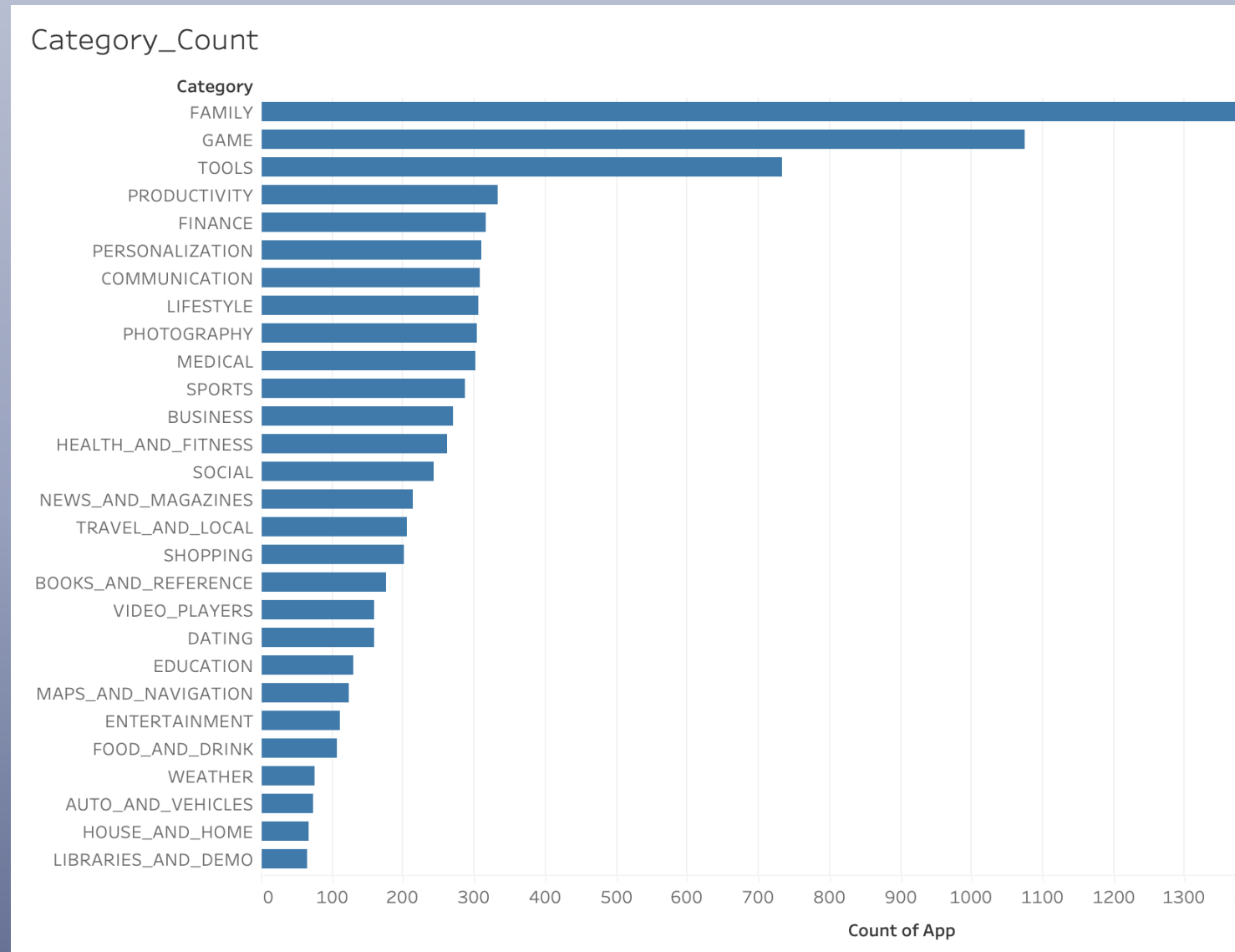
DATA AND TOOLS USED

- Dataset sourced from Kaggle (Google Play Store + User Reviews).
- Two datasets (apps + reviews) were merged using VLOOKUP and joined in Tableau.
- Additional data such as marketing spend or in-app purchase behavior would enhance prediction accuracy.
- Tools used: Excel (cleaning), Tableau (visualization).

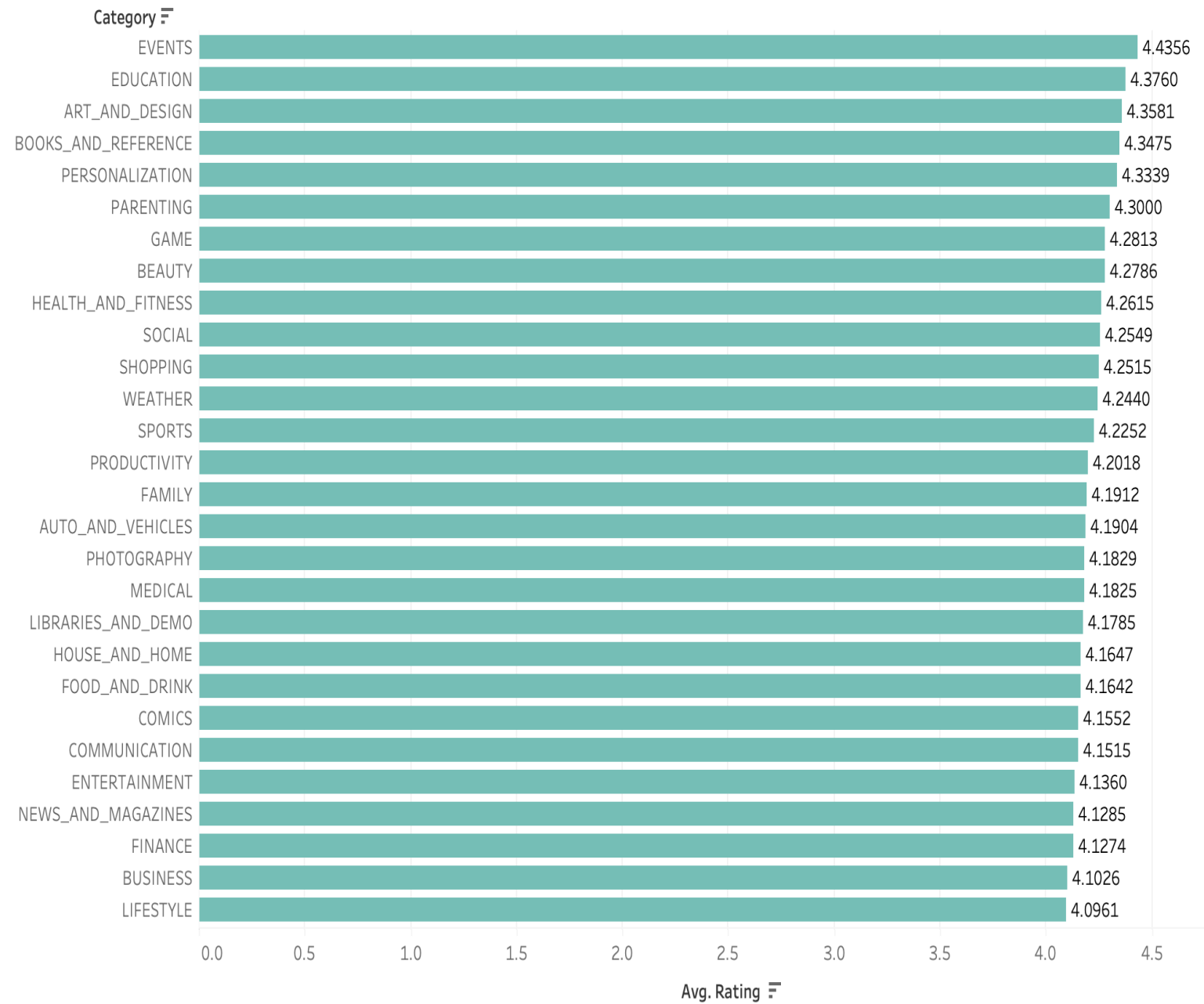
CATEGORY COUNT

The distribution of apps across categories shows significant variation.

Some categories, such as Family, Game, Tools, contain a very large number of apps, making them highly competitive. Meanwhile, smaller categories may offer less competition and potential opportunities for new apps



Category_Rating



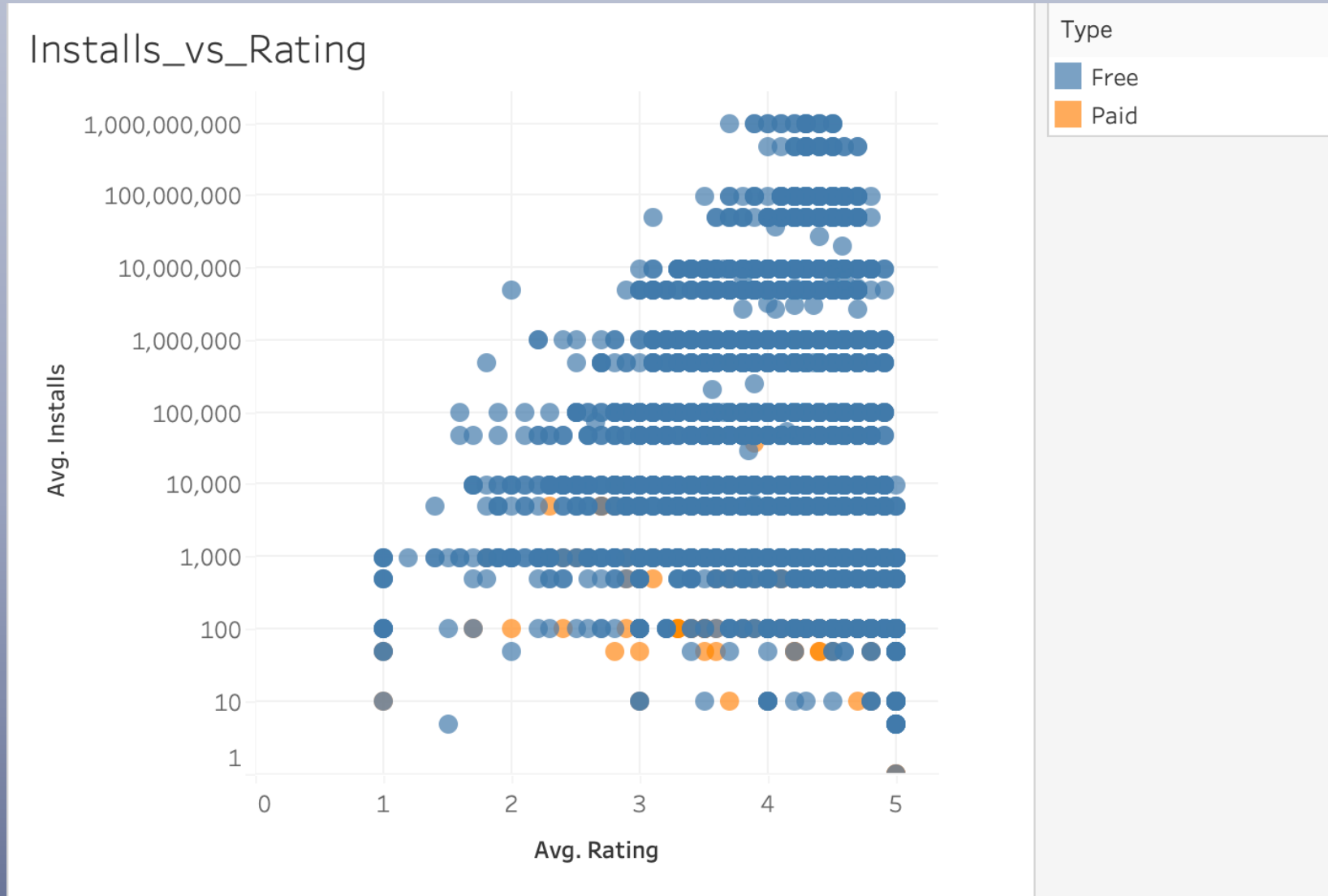
MARKET OVERVIEW (EDA).

Users on Google Play are generous—most categories score above 4 stars! A few standouts like Events and Education shine at the top, while others such as Lifestyle and Business lag slightly behind. Overall, the market shows strong satisfaction, but also clear spaces where better app design could win big.

INSTALLS VS RATING

Higher-rated apps generally achieve significantly more installs, showing a positive relationship between user satisfaction and popularity.

However, a few apps with average ratings still attract massive installs, suggesting that factors like brand awareness, category type, or marketing also influence success.

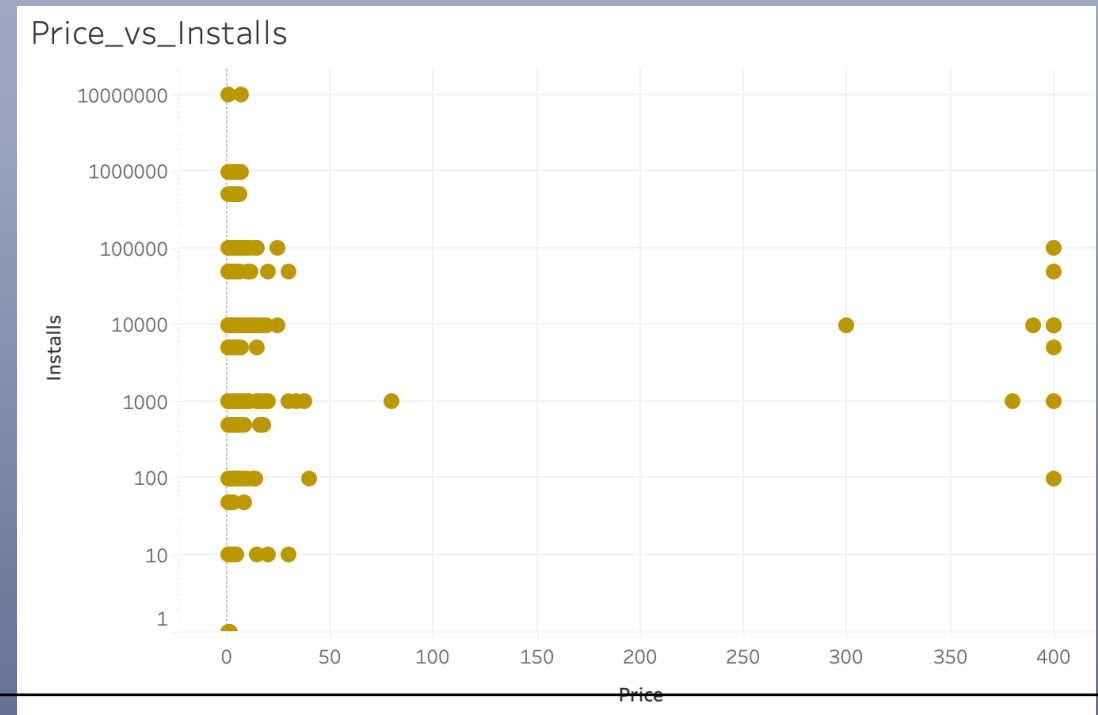
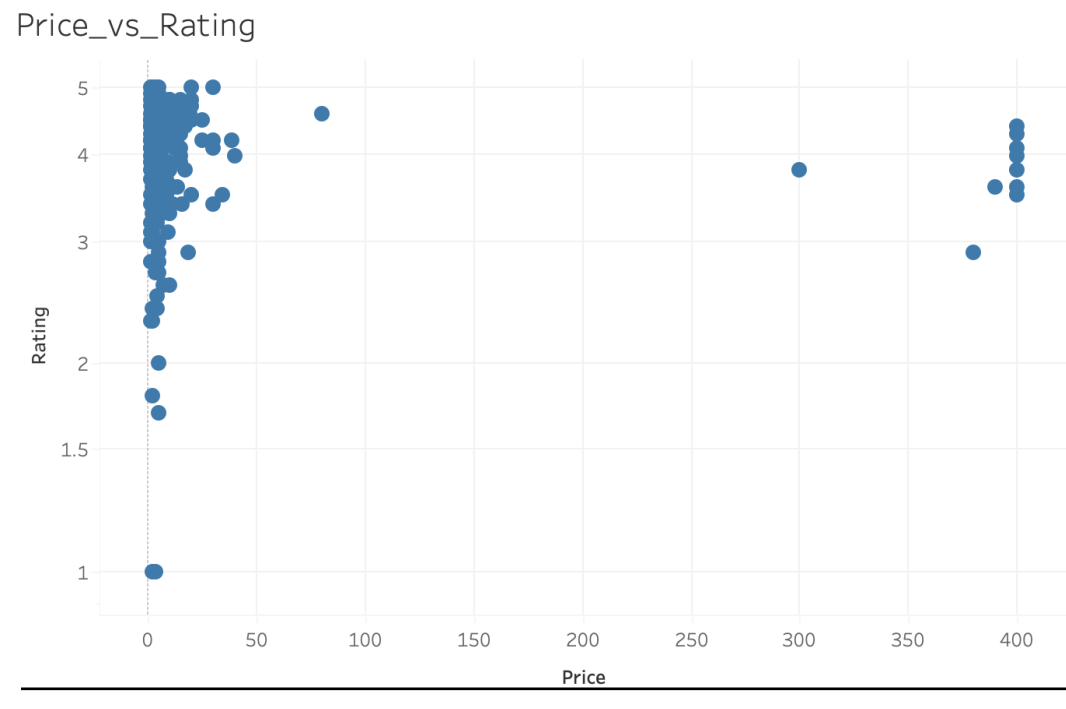


IMPACT OF PRICE ON APP SUCCESS

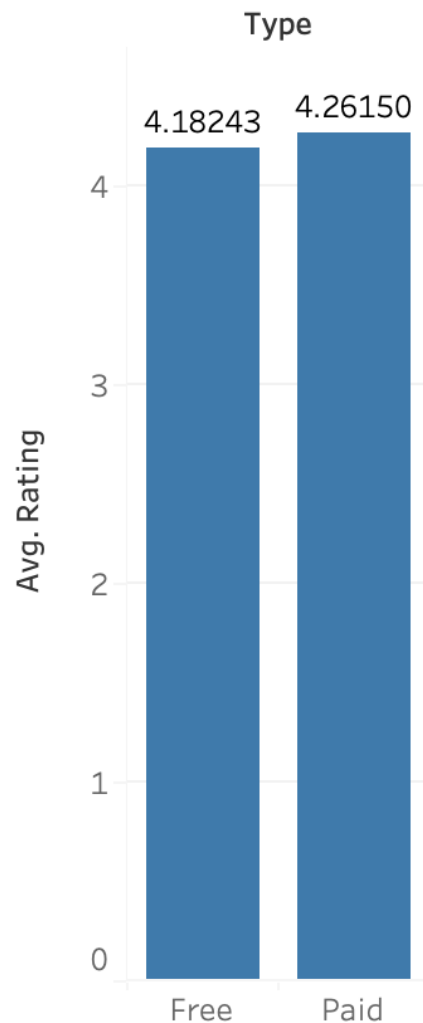
Price has limited influence on user ratings, but a strong impact on downloads.

Paid apps receive slightly higher ratings, yet their install volume drops sharply as prices increase.

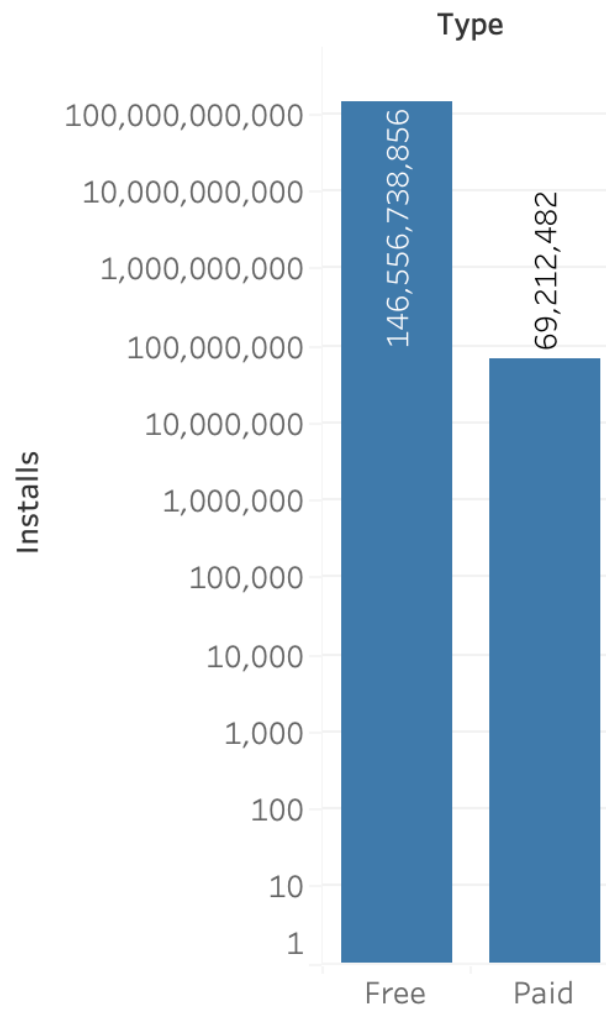
Overall, the Google Play market is highly price-sensitive, with free apps dominating



Rating_Free_vs_Paid



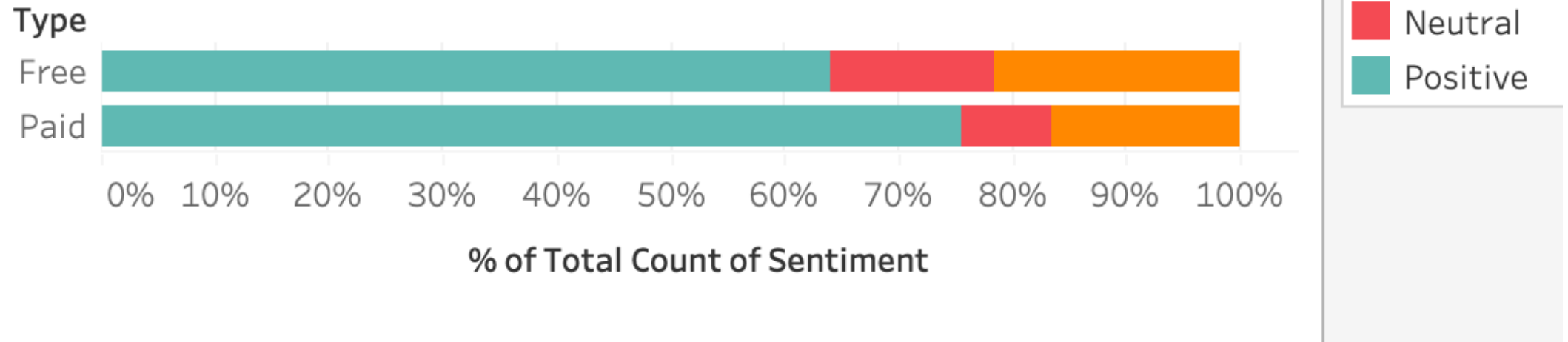
Installs_Free_vs_Paid



FREE VS PAID: PERFORMANCE COMPARISON

- Free apps dominate the market in terms of reach and installations. Paid apps, however, maintain slightly higher average ratings, indicating higher perceived quality.

Sentiment_Free_vs_Paid



SENTIMENT: FREE VS PAID APPS

Paid apps show a higher ratio of positive reviews but receive fewer total reviews.

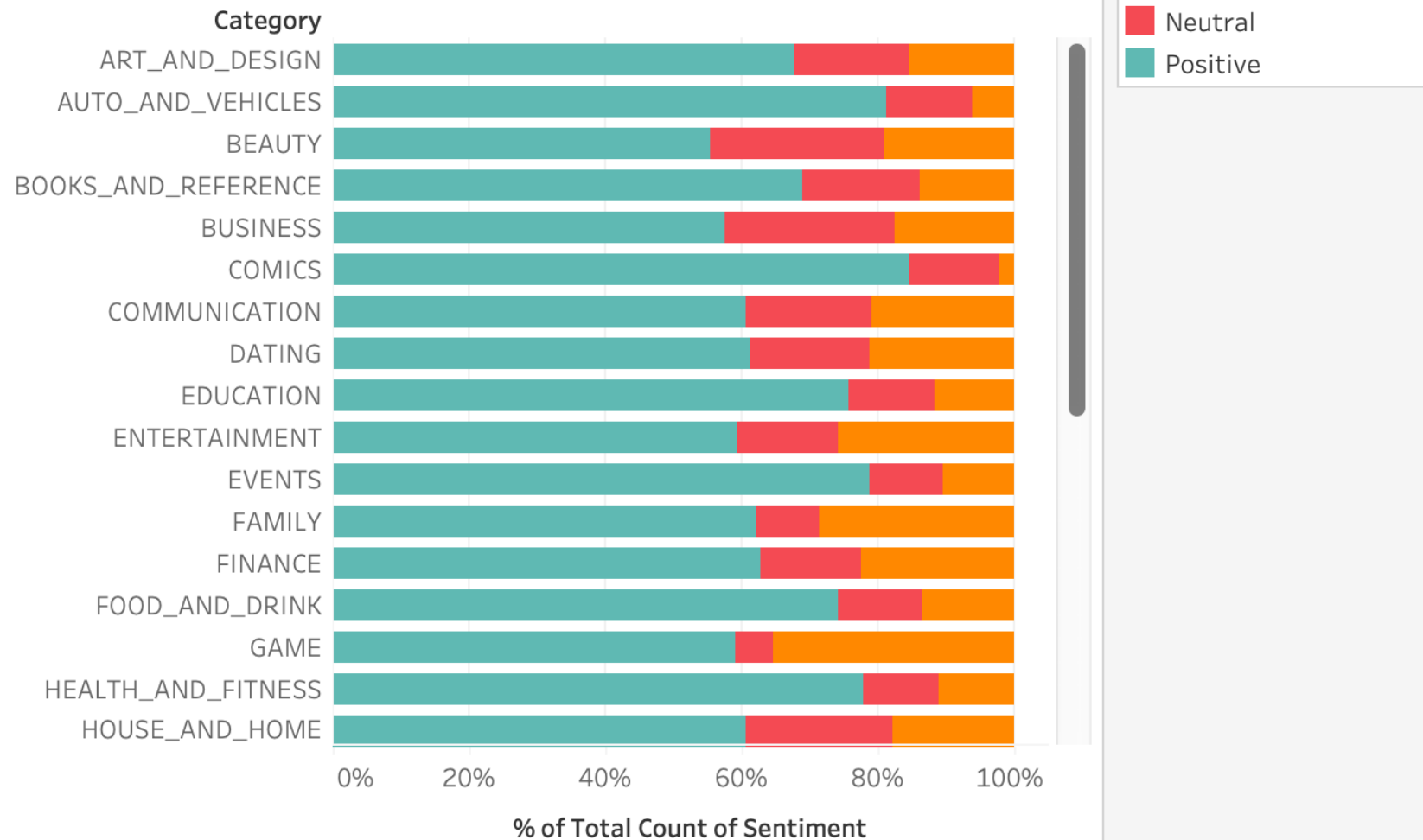
Free apps collect more negative feedback due to larger user bases and broader expectations.

SENTIMENT BY CATEGORY

Sentiment analysis reveals significant variation across categories. Some categories face higher negative review volume, signalling potential stability issues. This usually points to issues such as bugs, poor usability, too many ads, or privacy concerns.

Others maintain consistently positive sentiment and stronger user loyalty.

Sentiment_by_Category



STATISTICAL ANALYSIS

I performed a few key analyses to understand what drives app success on Google Play:

- **Correlation Analysis:**
To see whether higher ratings actually lead to more installs.
 - **Category Comparison:**
To identify which categories are crowded and which ones still have opportunities.
 - **Free vs Paid Analysis:**
To understand how pricing affects installs and ratings.
 - **Sentiment Analysis:**
To measure how users feel about apps in each category.
 - **These analyses together helped reveal the main performance patterns in the market.**
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CONCLUSION

App success on Google Play is shaped by a combination of user satisfaction, pricing strategy, and category dynamics.

Higher-rated apps generally attract more installs, but strong marketing or category demand can also drive high adoption.

While Free apps dominate downloads, Paid apps deliver higher satisfaction and stronger sentiment.

Understanding these patterns can help developers design better products and make smarter market decisions.



THANK YOU

APPENDIX

- Kaggle — Google Play Store Apps Dataset
- <https://www.kaggle.com/datasets/whenamancodes/play-store-apps>